

DEEP LEARNING — 046211
FINAL EXAM A
24/2/2025

EXAM DURATION: 3 HOURS

INSTRUCTIONS.

1. You can use a calculator and the 2 formula sheets (4 pages).
2. You need to answer all question parts, unless you received a special permit.
3. The weight of each question is written in the form. The total number of points is 100 points.
4. Please write clearly and orderly. You can write either in English or in Hebrew.
5. You must give the full details of the solution derivation. Solutions without explanations will not be given a score, unless written otherwise.
6. Once you finish the exam, you need to use Gradescope to upload your exam. In addition, you need to submit the notebook.

1. INVARIANCE AND EQUIVARIANCE IN LINEAR MODELS

Consider a linear regression problem with a neural network of the form $\hat{y} = \mathbf{x}^T \mathbf{w}$, where $\mathbf{x}, \mathbf{w} \in \mathbb{R}^d$. Assume the network is trained to minimize

$$\mathcal{L}_0(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (y^{(n)} - \mathbf{w}^T \mathbf{x}^{(n)})^2 = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2,$$

where $\mathbf{X} = [\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)}]^T$ is the sample set and $\mathbf{y} = [y^{(1)}, \dots, y^{(N)}]^T$ is the corresponding label vector.

We define operator $\mathbf{G}$ as some operator on the input $\mathbf{x}$, e.g. the input is an image and $\mathbf{G}$ spins it clockwise 90° . $\mathbf{G}$ is defined via an **invertible** $d \times d$ square matrix.

We define a function f as **invariant** w.r.t. $\mathbf{G}$ if $f(\mathbf{G}\mathbf{x}) = f(\mathbf{x})$, i.e. applying $\mathbf{G}$ does not change the output of f .

We will mark $\mathcal{G}$ a set of operators, and consider functions that are invariant w.r.t. all operators in $\mathcal{G}$. For example, if $\mathcal{G}$ is a set of clockwise rotation operators, an invariant f might be the sum of all pixel values.

The following are a set of properties that apply to rotation matrices (operators):

1. If $\mathbf{G} \in \mathcal{G}$, $\mathbf{G}^T \in \mathcal{G}$.
2. Any $\mathbf{G} \in \mathcal{G}$ is **orthogonal** ($\mathbf{G}\mathbf{G}^T = \mathbf{G}^T\mathbf{G} = \mathbf{I}$).
3. Applying operator $\mathbf{G} \in \mathcal{G}$ on all elements of $\mathcal{G}$ returns $\mathcal{G}$, meaning $\bigcup_{\mathbf{G}' \in \mathcal{G}} \mathbf{G}\mathbf{G}' = \mathcal{G}$.

We define the following cost function

$$\mathcal{L}_1(\mathbf{w}) = \frac{1}{2} \sum_{\mathbf{G} \in \mathcal{G}} \|\mathbf{y} - \mathbf{X}\mathbf{G}^T \mathbf{w}\|^2$$

- (1) Explain intuitively why this cost function encourages the predictor $\mathbf{w}$ to be invariant.
- (2) Show that $\mathcal{L}_1(\mathbf{w})$ is invariant to rotation transformations of the form $\mathbf{w} \rightarrow \mathbf{G}\mathbf{w}$, for all $\mathbf{G} \in \mathcal{G}$.
- (3) Prove that the optimal $\mathbf{w}_1^*$ that minimizes $\mathcal{L}_1(\mathbf{w})$ is

$$\mathbf{w}_1^* = \mathbf{K}^{-1} \mathbf{R} \mathbf{X}^T \mathbf{Y},$$

where $\mathbf{K} = \sum_{\mathbf{G} \in \mathcal{G}} \mathbf{G}^T \mathbf{X}^T \mathbf{X} \mathbf{G}$, $\mathbf{R} = \sum_{\mathbf{G} \in \mathcal{G}} \mathbf{G}$ and we assume $\mathbf{K}$ is invertible.

- (4) Show that $\mathbf{G}\mathbf{R} = \mathbf{R}$ for all $\mathbf{G} \in \mathcal{G}$.
- (5) Show that $\mathbf{G}\mathbf{K}^{-1}\mathbf{G}^T = \mathbf{K}^{-1}$ for all $\mathbf{G} \in \mathcal{G}$.
Hint: for invertible $\mathbf{A}, \mathbf{B}$, $(\mathbf{A}\mathbf{B})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$
- (6) Prove that the optimal $\mathbf{w}_1^*$ yields a predictor $\hat{y}(\mathbf{x}) = \mathbf{x}^T \mathbf{w}_1^*$ that is invariant to $\mathbf{x} \rightarrow \mathbf{G}\mathbf{x}$ for all $\mathbf{G} \in \mathcal{G}$.

Suppose that the network is now of the form $\hat{\mathbf{y}} = \mathbf{W}\mathbf{x}$, where $\mathbf{x} \in \mathbb{R}^d$, $\mathbf{W} \in \mathbb{R}^{d \times d}$, $\hat{\mathbf{y}} \in \mathbb{R}^d$.

A function f is **equivariant** to operator $\mathbf{G}$ if $f(\mathbf{G}\mathbf{x}) = \mathbf{G}f(\mathbf{x})$, i.e. applying the operator on the input is the same as applying the operator on the output.

Let $\mathbf{X} = [\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)}]^T$ be the sample set and $\mathbf{Y} = [\mathbf{y}^{(1)}, \dots, \mathbf{y}^{(N)}]^T$ is the corresponding label matrix.

- (7) Suggest a cost function $\mathcal{L}_2(\mathbf{w})$ that encourages the predictor $\hat{\mathbf{y}} = \mathbf{W}\mathbf{x}$ is equivariant for all $\mathbf{G} \in \mathcal{G}$. Provide a short explanation.

2. BATCH SIZE AND STEP SIZE IN SGD

Let $\mathcal{L}$ be a loss function which can be decomposed into N parts $\ell^{(n)}$

$$\mathcal{L}(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N \ell^{(n)}(\mathbf{w}) .$$

We examine minibatch-stochastic gradient descent training

$$\hat{\mathbf{g}}_t = \frac{1}{B} \sum_{n \in \mathcal{B}} \nabla \ell^{(n)}(\mathbf{w}_t)$$

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \hat{\mathbf{g}}_t$$

where $\hat{\mathbf{g}}_t$ is gradient estimate on the minibatch, $\mathcal{B}$ is a randomly selected subset of $\{1, \dots, N\}$ of size B , and η is the step size. We wish to understand how to select the step size η given the batch size B . For simplicity, we approximate the loss change after a single step using a second-order Taylor function

$$(2.1) \quad \mathcal{L}(\mathbf{w}_t - \eta \hat{\mathbf{g}}_t) \approx \mathcal{L}(\mathbf{w}_t) - \eta \hat{\mathbf{g}}_t^\top \mathbf{g}_t + \frac{1}{2} \eta^2 \hat{\mathbf{g}}_t^\top \mathbf{H}_t \hat{\mathbf{g}}_t$$

where $\mathbf{g}_t = \nabla \mathcal{L}(\mathbf{w}_t) = \mathbb{E} \hat{\mathbf{g}}_t$ and $\mathbf{H}_t = \nabla^2 \mathcal{L}(\mathbf{w}_t)$.

- (1) For full batch gradient descent, i.e. $B = N$, what is the optimal step size η which minimize the loss in Eq. 2.1?
- (2) Show that for a general batch size

$$(2.2) \quad \mathbb{E} \mathcal{L}(\mathbf{w}_t - \eta \hat{\mathbf{g}}_t) = \mathcal{L}(\mathbf{w}_t) - \eta a_t + \frac{1}{2} \eta^2 \left(b_t + \frac{c_t}{B} \right)$$

and write a_t, b_t and c_t explicitly as functions of $\mathbf{g}_t, \mathbf{H}_t$ and $\Sigma_t = \mathbb{E}(\hat{\mathbf{g}}_t \hat{\mathbf{g}}_t^\top) - \mathbf{g}_t \mathbf{g}_t^\top$, i.e. the covariance matrix of $\hat{\mathbf{g}}_t$. Hint: for any vector $\mathbf{v}$ and matrix $\mathbf{A}$, we have

$$\mathbf{v}^\top \mathbf{A} \mathbf{v} = \text{Tr}[\mathbf{v}^\top \mathbf{A} \mathbf{v}] = \text{Tr}[\mathbf{A} \mathbf{v} \mathbf{v}^\top] .$$

- (3) Is it feasible to calculate a_t, b_t and c_t directly? Suggest another method to efficiently calculate a_t, b_t and c_t without using second-order derivatives.

For simplicity, from now on, we assume that: $a_t = a, b_t = b$ and $c_t = c$ are constant.

- (4) What is the optimal step size η_* which minimizes the loss in Eq. 2.1? Does this approximately match what we saw in class empirically (consider two cases: very large B and very small B)? Explain.
- (5) What is the loss decrease after a taking a step with the optimal step size, i.e. $\mathbb{E} \mathcal{L}(\mathbf{w}_t - \eta_* \hat{\mathbf{g}}_t) - \mathcal{L}(\mathbf{w}_t)$? Does this approximately match what we saw in class empirically (consider two cases: very large B and very small B)? Explain.

3. MASKING IN TRANSFORMERS

We examine the classic encoder-decoder transformer model for next-token prediction, and focus on its masking mechanism.

- (1) Draw a picture of the transformer, and qualitatively explain all its parts.
- (2) Why do we need “Masked self-attention” in the decoder, instead of the regular self attention? Give an example with a specific task.
- (3) To implement Multi-Head Masked Self-Attention (MHMSA) in the decoder we use

$$\forall i \in \{1, \dots, H\} : \mathbf{Q}_i = \mathbf{XW}_i^Q, \mathbf{K}_i = \mathbf{XW}_i^K, \mathbf{V}_i = \mathbf{XW}_i^V$$

$$\text{MHMSA}(\mathbf{X}) = \sum_{i=1}^H \text{Softmax} \left(\frac{\mathbf{Q}_i \mathbf{K}_i^\top}{\sqrt{d}} + \mathbf{M} \right) \mathbf{V}_i \mathbf{W}_i^O \in \mathbb{R}^{T \times d}.$$

Explain how should we choose $\mathbf{M} \in \mathbb{R}^{T \times T}$ to implement masking based on the expression for MHMSA above, and why this works.

- (4) Why do we not require masking in the Multi-Head Self-Attention (MHSA) right after the MHMSA? Base this on the expression for MHSA.
- (5) Explain the difference in operation of the MHMSA in training and inference. Which method is faster?
- (6) A researcher thought it might be useful if (similarly to vision models) we add a “max-pooling” with pooling size 2 layer on the token dimension at the end of the first decoder blocks (after the fully connected layers), to reduce its size from T to $T/2$, and then extend it back to the original T (e.g., by replicating the tokens) before the last decoder block. Explain why is this problematic (hint: it is related to the reason we need masking), and how to fix the issue above, allowing this max-pooling layer to be added.